

BEDDING AND PIPE LOADING

I. HS-25 AND E-30 LOADING DIRECTLY BELOW THE CENTERLINE AT SHALLOW COVER (30 INCHES).

LIVE LOADS ARE THE CONTROLLING FACTOR IN THIS INSTALLATION. THIS SCENARIO IS ONLY VALID FOR THE 18 INCH LATERAL PIPES. DESIGN CALCULATIONS IN SECTION II VERIFY THAT A NATIVE MATERIAL MEETING AN ASTM D2487 CLASSIFICATION OF TYPE III STRUCTURAL BACKFILL (OR BETTER) IS AN ACCEPTABLE BACKFILL MATERIAL. COMPACTION OF 95% OR GREATER IS REQUIRED TO ACHIEVE AN E' VALUE OF 2000 PSI. UNDISTURBED NATIVE, IMPERVIOUS MATERIAL SHALL BE USED FOR THE BEDDING. WHEN SOFT OR YIELDING MATERIAL IS ENCOUNTERED IN THE TRENCH BOTTOM, THIS MATERIAL MUST BE REMOVED AND REPLACED WITH A MINIMUM 4 INCH LAYER OF CRUSHED STONE BEDDING. BACKFILL SHOULD EXTEND UP TO AND OVER THE CROWN OF THE PIPE BY 6 INCHES. TRENCH WIDTH SHALL BE NO LESS THAN 42 INCHES FOR THE 18 INCH DIAMETER PIPE.

II. HS-25 AND E-30 LOADING OFFSET 6 FT (MIN) FROM CENTERLINE.

THE SIX FOOT SEPARATION OF PIPE CENTERLINE AND RAIL CENTERLINE ALLOW FOR THE LIVE LOADS TO INTERACT IN A LESS PREDOMINANT MANNER THAN WHEN SITUATED DIRECTLY ABOVE THE PIPE CROWN. SINCE THE LIVE LOAD PROJECTS DOWNWARD AND OUTWARD THERE ARE NO LIVE LOADS PRESENT WHEN THE PIPE IS BURIED AT 30 INCHES DEPTH AND 6 FEET AWAY FROM THE RAIL CENTERLINE. AASHTO SECTION 18 ALLOWS FOR THE LIVE LOAD TO BE COMPLETELY DISREGARDED WHEN THE BURIAL IS BELOW 8 FEET SO WHEN AT 10 OR 20 FEET OF DEPTH NO LIVE LOADS ARE PRESENT. DESIGN CALCULATIONS IN SECTION II VERIFY THAT A NATIVE MATERIAL MEETING AN ASTM D2487 CLASSIFICATION OF TYPE III STRUCTURAL BACKFILL (OR BETTER) IS AN ACCEPTABLE BACKFILL MATERIAL. COMPACTION OF 95% OR GREATER IS REQUIRED TO ACHIEVE AND E' VALUE OF 2000 PSI. UNDISTURBED NATIVE, IMPERVIOUS MATERIAL SHALL BE USED FOR THE BEDDING. WHEN SOFT OR YIELDING MATERIAL IS ENCOUNTERED IN THE TRENCH BOTTOM, THIS MATERIAL MUST BE REMOVED AND REPLACED WITH A MINIMUM 4 INCH LAYER OF CRUSHED STONE BEDDING. BACKFILL SHOULD EXTEND UP TO AND OVER THE CROWN OF THE PIPE BY 6 INCHES. TRENCH WIDTH SHALL BE NO LESS THAN 42 INCHES FOR THE 18-INCH DIAMETER PIPE AND 48 INCHES FOR THE 24-INCH DIAMETER PIPE. (TRENCH WIDTH MAY NEED TO BE GREATER TO ACHIEVE THE DESIRED COMPACTION.) WHEN AT GREATER THAN FIVE FEET OF DEPTH A TRENCH BOX MAY HAVE TO BE USED. IF A TRENCH BOX IS USED CARE MUST BE TAKEN WHEN REMOVING THE TRENCH BOX SO AS TO NOT DISTURB THE PIPE ZONE DURING ITS REMOVAL.

• HANCOR PROPRIETARY ANALYSIS REPORTED DATED 10-24-03. BY

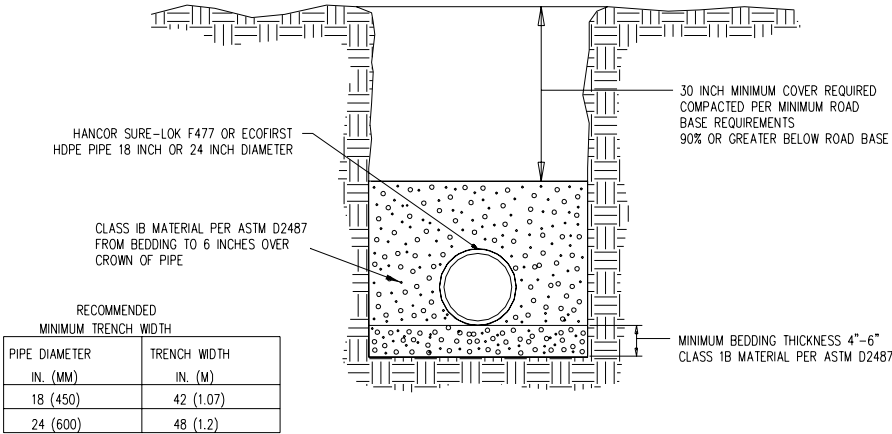
BEDDING & SLURRY WITHIN CITY STREETS

1. ALL STORM DRAIN PIPE BEDDING WITHIN CITY OF LOS ANGELES STREETS SHALL BE CONSTRUCTED IN ACCORDANCE WITH CITY STD PLAN S-251-1, CASE 2.
2. BACKFILL WITH SLURRY PER SSPWC, 100-E-100 TO BOTTOM OF PAVEMENT SECTION WITHIN CITY STREETS.

VENDOR DRAWING

FOR REFERENCE ONLY

18" HDPE TRENCH DETAILS
CENTERLINE – IMPORTED ROCK



REFERENCE ASTM D2321, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PIPE FOR SEWERS AND OTHER GRAVITY-FLOW APPLICATIONS."

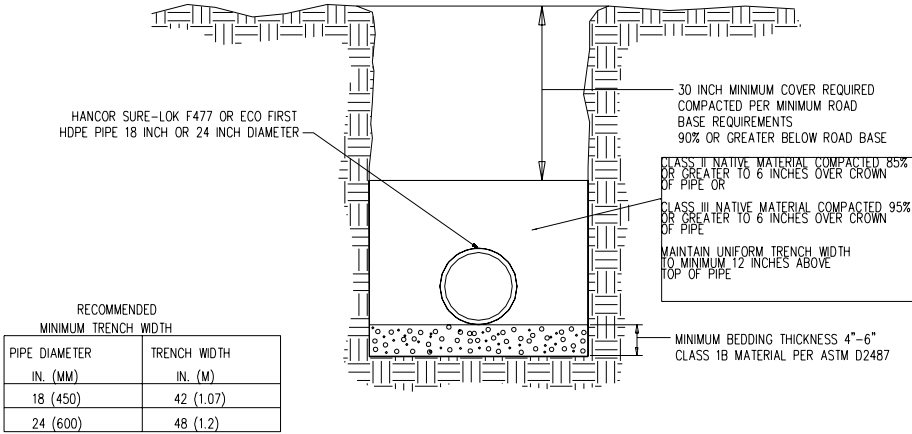


FINDLAY, OHIO 45839

HDPE TRENCH INSTALLATION DETAIL

SCALE: NTS
DATE: 10-30-2003
DRN BY: MDK

18" / 24" HDPE TRENCH DETAILS
OFF CENTERLINE – NATIVE MATERIAL



REFERENCE ASTM D2321, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PIPE FOR SEWERS AND OTHER GRAVITY-FLOW APPLICATIONS."

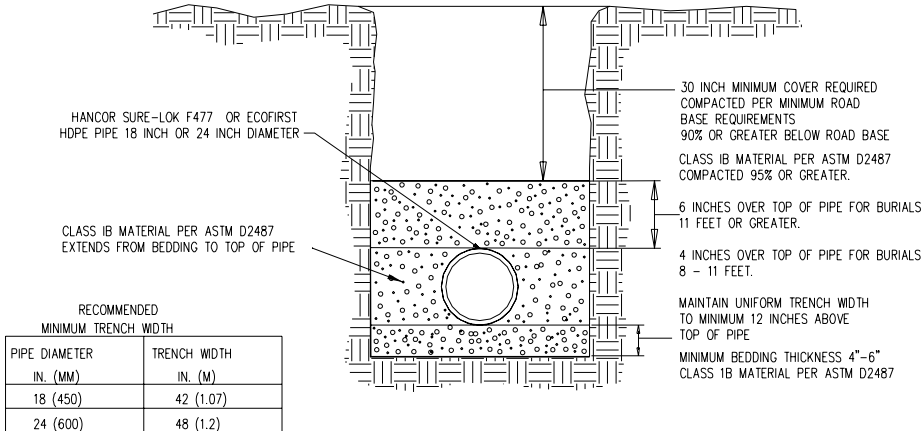


FINDLAY, OHIO 45839

HDPE TRENCH INSTALLATION DETAIL

SCALE: NTS
DATE: 10-30-2003
DRN BY: MDK

18" / 24" HDPE TRENCH DETAILS
OFF CENTERLINE – IMPORTED ROCK – 8 FEET OR GREATER



REFERENCE ASTM D2321, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PIPE FOR SEWERS AND OTHER GRAVITY-FLOW APPLICATIONS."

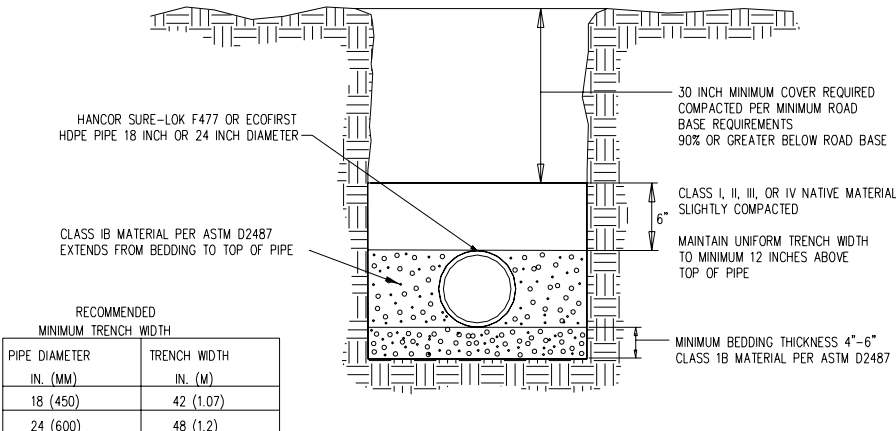


FINDLAY, OHIO 45839

HDPE TRENCH INSTALLATION DETAIL

SCALE: NTS
DATE: 10-30-2003
DRN BY: MDK

18" / 24" HDPE TRENCH DETAILS
OFF CENTERLINE – IMPORTED ROCK – 8 FEET OR LESS



REFERENCE ASTM D2321, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PIPE FOR SEWERS AND OTHER GRAVITY-FLOW APPLICATIONS."



FINDLAY, OHIO 45839

HDPE TRENCH INSTALLATION DETAIL

SCALE: NTS
DATE: 10-30-2003
DRN BY: MDK



LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
SAN FERNANDO VALLEY EAST-WEST BRT PROJECT



SHIMMICK
OBAYASHI, JV
5805 SEPULVEDA BLVD., SUITE 500
VAN NUYS, CA 91411

BUSWAY SEGMENT 1 EAST
DRAINAGE DETAILS

SHEET 3 OF 6

CONTRACT NO	C0675
DRAWING NO	BIC6035
SCALE	NO SCALE
SHEET NO	